GLS University
Faculty of Computer Applications & IT
iMSCIT SEM 6
221601605 Practicals on Machine Learning
Unit 4
Practical Assignment

1
- Generate 300 samples with 4 centers using `make_blobs()`.
- Apply K-means with `n_clusters=4` and visualize the clustered data with the centroids shown in red.

2
- Use Iris dataset with 150 samples and 4 features (use only the first two features: Sepal Length and Sepal Width).
- K-means clusters the data into 3 clusters, corresponding to the 3 species in the Iris dataset.

3 Create a dataset as below:

```
data = {
  'CustomerID': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
   'Annual Income (k$)': [15, 16, 17, 18, 19, 10, 14, 12, 13, 11],
   'Spending Score (1-100)': [39, 81, 6, 77, 40, 3, 48, 49, 47, 42]
}
```

Implement K-means to segment customers into 3 groups based on their annual income and spending score.

4 Use `load_digits()` function gives a dataset of 8x8 pixel images of handwritten digits.
- Apply K-means clustering with `n_clusters=10` since there are 10 digits (0-9).
- Visualize the cluster centers, which represent the average "digit" image for each cluster.

5 Create following dataset:

```
data = {
   'City': ['New York', 'Los Angeles', 'Chicago', 'Houston', 'Phoenix', 'Philadelphia', 'San Antonio', 'San Diego', 'Dallas', 'San Jose'],
   'Latitude': [40.7128, 34.0522, 41.8781, 29.7604, 33.4484, 39.9526, 29.4241, 32.7157, 32.7767, 37.3382],
   'Longitude': [-74.0060, -118.2437, -87.6298, -95.3698, -112.0740, -75.1652, -98.4936, -117.1611, -96.7970, -121.8863]
}
```

- Apply K-means clustering to group cities based on their geographical location.
- Visualize the cities and the centroids of the clusters on a map-like plot.